

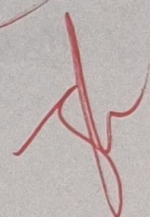
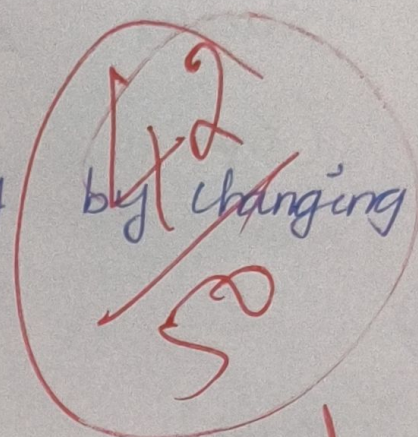
Analytical problem

1.) Grey Level Transformation:-

a.) Purpose:

It is used to improve image quality pixel intensity value.

- (i) Enhancing Contrast.
- (ii) Highlighting image
- (iii) making dark region brighter
or
bright region darker.
- (iv) Improves visibility for analysis.



b.) Negative Transformation:

formula: $S = 255 - r$

Original Pixel (r)	50	100	150	200
Transformed Pixel (s)	205	155	105	55

- Dark pixel $\rightarrow$ brighter
- Bright pixel $\rightarrow$ darker
- This images appears negative (inverse)

2.) Histogram Equalization:-

a.) Used:

- It improves image contrast
- It shows hidden details in low contrast image
- It spreads the pixels across the available intensity range.

b) Intensity	0	1	2	3
frequency	4	6	10	6
CDF	4	10	20	26

c)

Formula:

$$s = \frac{CDF}{26} \times (4-1)$$

equalized value

(i) $(4/26) \times 3 = 0.46$

~~equal~~

↓
0

(ii) $(10/26) \times 3 = 1.15$

1

(iii) $(20/26) \times 3 = 2.31$

2

(iv) $(26/26) \times 3 = 3$

3

⇒ The intensity value is distributed across the range resulting in better image.

3) Gaussian Smoothing:-

- a)
- (i) reduce image noise smoothing
 - (ii) preserves edges better than averaging.
 - (iii) gives more weight to center pixel.

b) Calculation:

$$= \frac{1}{16} (10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10)$$

$$= \frac{360}{16} = 22.5 //$$

$$\text{Value} = 22.5 \rightarrow 23 //$$

$\Rightarrow$ The center pixel from 40 $\rightarrow$ 23 reduce noise and smoother images.

Edge Detection :-

identify the sudden changes in intensity

detects $\rightarrow$ (i) object boundaries

(ii) shapes.

(iii) features changes

Gradient :-

$$\Rightarrow (-1 \times 10) + (-2 \times 10) + (-1 \times 10) + 0 + 0 + 0 (1 \times 30) \\ + (2 \times 30) + (1 \times 30)$$

$$= -10 - 20 - 10 + 30 + 60 + 30$$

$$= 80 //$$

$\Rightarrow$ Gradient = 80 indicate strong horizontal edges.

5.) Contrast Stretching :-

- a.)
- increases image contrast
 - Expands a narrow intensity range
 - makes details clearer.

b.)

Formula:

$$s = \frac{(r - 50) \times 255}{150 - 50}$$

$$r = 50, \quad s = \frac{(50 - 50) \times 255}{100} = 0 //$$

$$r = 100, \quad s = \frac{(100 - 50) \times 255}{100} = 127.5 \Rightarrow 128 //$$

$$r = 150, \quad s = \frac{(150 - 50) \times 255}{100} = 255 //$$

original (r)	50	100	150
new value (s)	0	128	255

$\Rightarrow$ It produces a higher-contrast image with improved visibility of details.